ABSTRACT

Core Size Dependence in Pulsed Laser-Induced Plasmonic Catalysis Observed in Gold@Silver Core@Shell Nanoparticle Synthesis

Student Name: Ryan Eriansyah

Student ID Number (NPM): 2206050365

Study Program: Undergraduate Physics

Plasmonic catalysis has emerged as a promising strategy for enhancing chemical reactions through localized surface plasmon resonance (LSPR) excitation in metallic nanoparticles. When illuminated at their plasmon resonance wavelength, noble metal nanoparticles such as gold and silver are capable of generating energetic hot carriers and localized photothermal heating, both of which can significantly influence catalytic reactions. Compared to conventional thermal catalysis, plasmonic systems offer several advantages, including localized energy conversion, improved reaction selectivity, and reduced bulk heating. Although numerous studies have investigated plasmonic catalysis under continuous-wave (CW) illumination, experimental studies utilizing pulsed laser excitation remain relatively limited. Recent theoretical works suggest that pulsed laser excitation may produce superior catalytic performance due to its extremely high peak power and transient thermal dynamics, enabling enhanced hot-electron generation while minimizing excessive heating. However, the influence of nanoparticle size on pulsed laser-induced plasmonic catalysis has not yet been systematically investigated. This study provides a new approach to understanding the role of nanoparticle core size in plasmon-driven catalytic reactions under nanosecond pulsed laser irradiation using gold@silver core@shell nanoparticle synthesis as a model catalytic system. The objective of this study is to investigate the effect of gold nanoparticle core size on silver shell growth kinetics. In this work, spherical gold nanoparticles were synthesized through a seed-mediated growth method and subsequently used as plasmonic cores for gold@silver core@shell formation. Five different gold nanoparticle samples with LSPR peaks around 520 nm, 525 nm, 532 nm, 536 nm, and 540 nm were successfully synthesized. Silver shell growth was induced using a nanosecond pulsed Nd:YAG laser operating at 532 nm with laser powers ranging from 50 mW to 400 mW. Real-time UV-Vis spectroscopy was employed to monitor the evolution of the LSPR peak during shell growth, while scanning electron

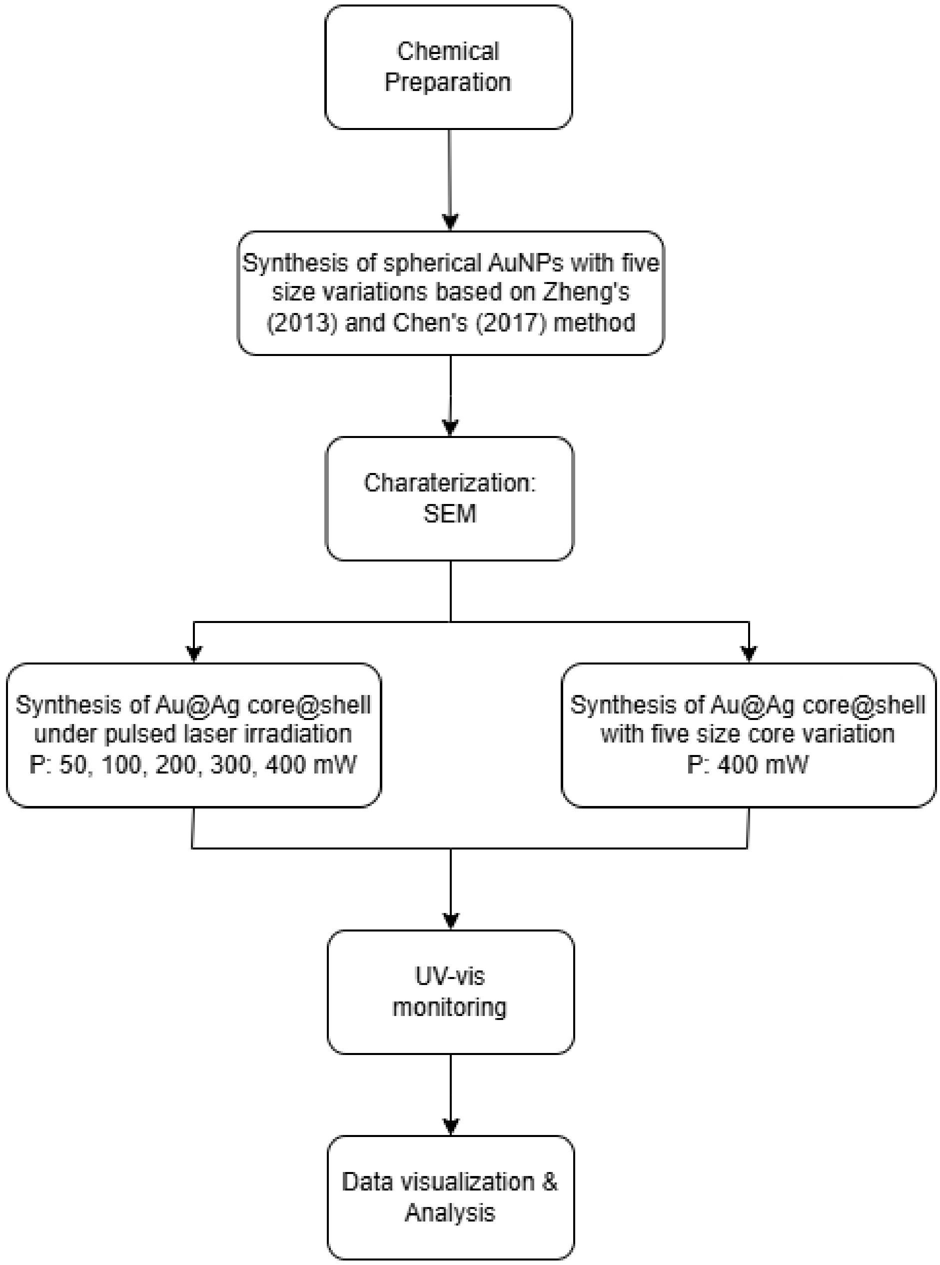

Figure 1. The schematic diagram of the research methodologies in the present work.

microscopy (SEM) was used to characterize nanoparticle morphology and shell formation behavior. To ensure the research workflow can be clearly understood, the schematic diagram of the research methodology is presented in **Figure 1**.

The experimental results demonstrate that the optical response of spherical gold nanoparticles strongly depends on nanoparticle size. Increasing the nanoparticle diameter caused a gradual red-shift of the LSPR peak accompanied by enhanced extinction intensity. Gold nanoparticles synthesized using the modified overgrowth method adapted from Chen et al., successfully produced LSPR peaks around 520 nm, 525 nm, 532 nm,

536 nm, and 540 nm, corresponding to different nanoparticle sizes. The observed red-shift is attributed to the increased oscillation path of conduction electrons within larger nanoparticles, which modifies the resonance condition of localized surface plasmons. However, at larger particle sizes, asymmetrical peak broadening was observed, particularly for the 540 nm sample, indicating deviations from ideal spherical morphology and the emergence of anisotropic growth behavior. These results reveal the importance of controlling nanoparticle size and morphology in plasmonic systems because even small variations in particle geometry can significantly alter the optical response and catalytic behavior. The real-time UV-Vis measurements further demonstrated that silver shell growth under pulsed laser irradiation resulted in a gradual blue-shift of the LSPR peak, confirming the formation of gold@silver core@shell structures. The blue-shift occurred because the plasmonic response became increasingly dominated by silver as shell thickness increased. The shell growth kinetics also showed a clear dependence on laser power and nanoparticle resonance matching with the excitation wavelength.

As a result, a clear trend of shell growth rate increasing with laser power was observed. Within Baldi's (2025) framework, where the dominant mechanism is d-band carriers, this result reverses the prevailing trend reported for spherical nanoparticles. Visual inspection of the LSPR spectra indicates that the synthesized nanoparticles are not perfectly spherical, which may explain why the trend is reversed. Non-spherical shapes have tips and corners where generated hot carriers are concentrated, accelerating the shell growth. Further confirmation via SEM is needed to verify the non-spherical morphology

The findings of this study provide important scientific implications for the design and optimization of plasmonic catalytic systems under pulsed laser excitation. First, the results demonstrate that nanoparticle size is a critical parameter governing plasmonic catalytic efficiency because the resonance condition directly influences hot-carrier generation and local electromagnetic field enhancement. Second, the study highlights the potential advantages of nanosecond pulsed laser excitation in plasmonic catalysis compared to conventional continuous-wave illumination. The transient high peak power generated during pulsed excitation may improve catalytic performance while minimizing excessive bulk heating and undesirable side reactions. Third, the successful correlation between experimental UV-Vis measurements and Mie theory simulations provides a useful strategy for predicting nanoparticle optical evolution during shell growth processes. These findings contribute to a deeper understanding of plasmon-mediated catalytic mechanisms and may enable more rational approaches for designing plasmonic nanomaterials with tunable optical and catalytic properties. Furthermore, the ability to precisely control shell growth through plasmonic excitation may be useful for applications in photocatalysis, nanoscale synthesis, sensing, and energy conversion technologies.

Keyword : gold nanoparticle, core@shell nanostructure, plasmonic, catalysis, localized surface plasmon resonance, pulsed laser

LIST OF CONTENTS

LIST OF FIGURES

CHAPTER 1
INTRODUCTION

1.1. Research Background

Catalyst plays a central role in today's chemical industrty. It is estimated that more than 80% of industrial chemical products rely on catalytic processes (Catlow et al., 2016). By lowering activation energy and improving reaction pathways, catalysts make many chemical processes more efficient, more economical, and more sustainable. However, despite their importance, many conventional catalytic systems still require energy-intensive operating conditions (Isahak & Al-Amiery, 2024). Conventional catalytic processes still rely on high temperature and high pressure, which raises concerns regarding energy consumptionn, operating costs, and environmental impact (Liu & Corma, 2023). These limitations have driven the development of catalytic approaches that can operate at lower energy requirements while maintaining high activity and high selectivity.

One promising direction is light-driven catalysis, where photons are used as an additional or alternative to trigger or accelerate chemical reactions. Among various light-based approaches, plasmonic catalysis has gained significant attention due to its ability to utilize the unique optical response of metal nanoparticles (Yuan et al., 2023). Plasmonic catalysis is based on the excitation of localized surface plasmon resonances (LSPR) in metal nanoparticles such as gold or silver. LSPR is a phenomenon where conduction electrons in metal nanoparticles collectively oscillate in response to the incident light (Baldi & Askes, 2023a). When LSPR is excited, metal nanoparticles exhibit intense local electromagnetic fields, efficient absorption of light energy, and strong light–matter interactions (Brongersma et al., 2015). As the excited plasmons decay, the relaxation of these excited plasmons can follow several pathways, including (i) the generation of energetic hot electrons and holes, and/or (ii) the conversion of optical energy into photothermal effect.

These plasmon-driven mechanisms can significantly influence catalytic reactions by accelerating reaction kinetics, modifying selectivity, or even enabling reactions that are difficult to achieve thermally (Kamarudheen et al., 2018). The plasmonics effect to enhance chemical reaction is fundamentally contrast to the conventional way, for example the ejected hot electrons can be transferred efficiently to reduce target reactants and the thermal effect is so-localized (around the nanoparticle), eliminating un-controlled bulk heating that induces subsequent unwanted side-products. Because of these advantages, plasmonic catalyst have been shown to promote reactions such as CO_2 reduction (Ou et al., 2023), hydrogen evolution (Strek et al., 2021), water splitting (Tack et al., 2024), and selective oxidation (Bizhani et al., 2024).

Despite the promise of plasmonics hot-electron and photothermal effects to enhance catalytic reaction, there was questions on which factors predominates: hot-electron OR photothermal? In 2018, Kamarudheen et al., (2018) elegantly established a model catalysis system to study the dominant factors between the two i.e. by in-situ observation of the growth of Ag shell on gold plasmonics-active nanoparticle. In this case they used CW laser to excite the plasmonics. They conclude that the branching is predominantly hot-electron driven-locally within the laser exposure zone and photothermal-away from the laser zone.

However, nearly all of these studies have relied on continous-wave (CW) light sources.To this date, most of the plasmonics-enhanced catalytic work is demonstrated using CW lightsource, while more advanced type of laser, i.e. pulsed laser, promise catalysis enhancement due to fundamentally contrast physics feature in comparison to CW kind. Pulsed-laser, in contrast to CW, is interesting because they are fundamentally different. Pulsed-laser is characterized by a repetitive on-off laser with peak power tremendeously higher to that of CW (despite of similar average power). The extra high peak power may eject gigantic number of hot-electron that subsequently improve the adjacent catalysis process while the intermittent off-period allows "time to breath" for the reaction products to desorbs from the catalyst interface to enchance the overall selectivity (Baldi & Askes, 2023)

Recently, Baldi and Askes (2023) performed a numerical comparison of plasmon-enhanced photothermal catalysis under CW and pulsed laser excitation. Their findings revealed that pulsed laser excitation can significantly increase product formation and selectivity as well to improve energy efficiency relative to CW illumination. Additionally, the transient high-temperature spikes generated during pulsed excitation were shown to mitigate catalyst poisoning by promoting desorption of species that block active catalytic sites. Despite these promising results, research on pulsed laser-induced plasmonic catalysis remains limited, especially from an experimental perspective. As a result, our understanding of how to effectively utilize pulsed laser illumination for plasmon-driven catalytic processes is still incomplete. This knowledge gap highlights the need for further investigation and presents a valuable opportunity for the development of more efficient plasmonic catalytic systems.

Inspired by these studies (Kamarudheen et al., 2018; Baldi & Askes, 2023;), the present research explores a model plasmonic catalytic system based on the synthesis of silver (Ag) shells on gold (Au) nanoparticle cores under nanosecond pulsed laser excitation. A previous thesis project by Pontoh (2025) demonstrated that nanosecond pulsed laser irradiation enhances silver shell formation more significantly than continuous-wave (CW) excitation, suggesting a role for either photothermal or hot electron effects.

However, Pontoh's study is limited only to single size only while it has been an established knowledge in this field that optical response and plasmonic strength of gold nanoparticles are strongly size-dependent, variations in core size are expected to significantly affect shell growth behavior of gold@silver core@shell nanoparticles. Futhermore, recently, Nguyen and Baldi (2025) reported a strong dependence of plasmonics nanoparticle size to hot electron production under CW-laser irradiation.

Therefore, as advancement of these foundational studies, this project is aimed to systematically vary the gold core size to investigate how pulsed laser excitation influences the balance between photothermal heating and hot electron injection during shell growth.

1.2. Problem Formulation

This study aims to answer following problems:

1. What is the effect of gold core size on the growth rate of silver shell under pulsed laser irradiation?

1.3. Reseach Objective

The study objective to be accomplished from this study are:

1. To study the effect of gold core size on the silver shell growth rate under pulsed laser irradiaton.

1.4. Research Benefit

This study offers a deeper understanding of how the size of the gold core influence plasmonic catalysis under nanosecond pulsed-laser excitation, addresing a knowledge gap in size-dependent plasmonic reaction mechanisms including the understanding of the dominant mechanism. The results enable more rational approach to designing and optimizing plasmon-driven synthesis, making it possible to control nanoparticle morphology and optical response more precisely.

1.5. Research Scope

The problem scope of this study is limited of the following statements:

1. This study limited to the synthesis of spherical gold@silver core@shell and does not include other shapes.
2. The excitation wavelength of the nanosecond pulsed laser is limited to 532 nm.
3. The primary variable studies is gold core size, while other synthesis parameters such as silver precursor concentration, reaction time, and solvent condition are kept constant.

CHAPTER 2
LITERATURE REVIEW

2.1. Localized Surface Plasmon Resonance (LSPR)

When the conduction electrons of metal nanoparticles interact with incoming light, this leads to oscillation of these electrons in the metal. The electrons are free to move within the boundaries of the particle but are localized at its surface. If the frequency of the oscillation of free electrons matches with the frequency of the incoming light, localized surface plasmon resonance (LSPR) is excited, as illustrated in **Figure 2.1** (Kelly et al., 2003).

If a nanoparticle is much smaller than the wavelength of incoming light, then all the electrons inside the nanoparticle experience the same electric field and oscillate in phase. This condition referred to as the quasi-static approximation (QSA). The excitation of LSPR leads to strong absorption cross-section (σ_{abs}) and/or scattering (σ_{scat}) by the metal nanoparticles, often surpassing their physical geometrical area. This effect producing a high extinction cross-section (σ_{ext}) for metal nanoparticles. The absorption, scattering, and extinction cross-section for metal nanoparticles can be approximated using the quasi-static model, as shown in Equations 1-3 (Enoch & Bonod, 2012).

$$\sigma_{\mathrm{scat}}(\omega) = \frac{2\pi^3}{\lambda^4} r^6 \left|\frac{\epsilon_m - \epsilon_d}{\epsilon_m + 2\epsilon_d}\right| \quad (1)$$

$$\sigma_{\mathrm{abs}}(\omega) = \frac{2\pi^3}{\lambda} r^3 \left|\frac{\epsilon_m - \epsilon_d}{\epsilon_m + 2\epsilon_d}\right| \quad (2)$$

Where λ is wavelength, r is the radius of the sphere, and ϵ_d and ϵ_{m} is the dielectric function of the nanoparticles and the dielectric function of the medium, respectively.

$$\sigma_{ext} = \sigma_{\mathrm{scat}}(\omega) + \sigma_{\mathrm{abs}}(\omega) \quad (3)$$

Therefore, for large particles, the extinction cross-section is primaly governed by scattering, which proportional to r^6 and for small nanoparticles it is mainly determined by absorption, proportional to r^3. Additionally, both scattering and absorption are influenced are strongly affected by the polarizability of the dielectric environment, especially in the case of spherical nanoparticles.

$$\alpha = 4\pi r^3 \left[\frac{\epsilon_m - \epsilon_d}{\epsilon_m + 2\epsilon_d}\right] \quad (4)$$

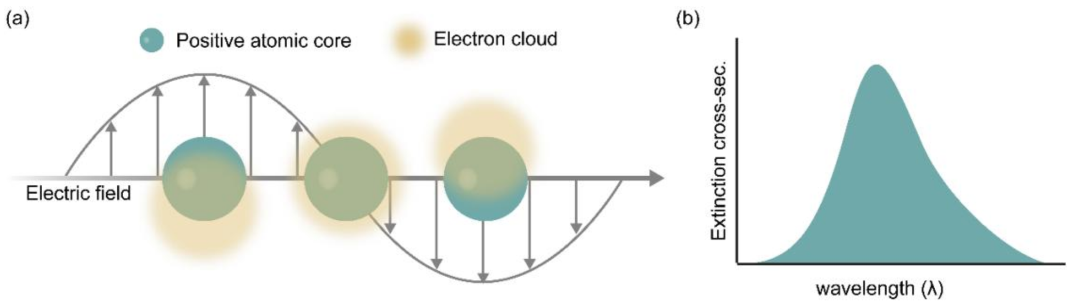

Figure 2.1. (a) Schematic illustration of localized surface plasmon resonance of metal nanoparticles. (b) In the far field, the resonance appears as a distinct peak in the extinction spectrum (Darmadi, I., 2021).

For spherical metall nanoparticles, the dielectric function $\epsilon_m(\omega)$ can be described by the Drude model, where it is represented in term its real and imaginary component as $\epsilon_m(\omega) = \epsilon_1(\omega) + i\epsilon_2(\omega)$.

The plasmon peak appears when the polarizability is maximized, which happens when the real part of the metal's dielectric function matches the condition $\mathrm{Re}[\epsilon(\omega)] = -2\epsilon_\mathrm{m}$ known as the Fröhlich resonance. From this condition, the analysis of the LSPR frequency gives $\omega_{LSPR} = \frac{\omega_p}{\sqrt{1+2\epsilon_d}}$, where ω_p is the plasma frequency, described as $\omega_p = \sqrt{\frac{nq^2}{m\epsilon_0}}$. This relationship explains where the LSPR wavelength will fall, and it also explains why the plasmon peak shifts whenever the refracative index of the surrounding medium changes (Mcoyi et al., 2024).

2.2. Effect of Nanoparticle Size on LSPR

Many studies have demonstrated that nanoparticles size plays a crucial role in determining their optical behavior, especially in relation to absorption and scattering cross-section. As shown in **Figure 2.2,** spherical gold nanoparticles with diameter ranging from 5 to 100 nm, the results reveal that increasing particles size initially enhances the absoprtion cross-section per particles volume, but beyond a certain size, the value begins to decrease (Ferrari, 2023).

In addition, an increase in nanoparticle size results in broadening of the absorption bandwidth and a slight red-shift in the LSPR peak. The strength of the electric dipole moment also increases, leading to stronger absorption cross-section. However, larger

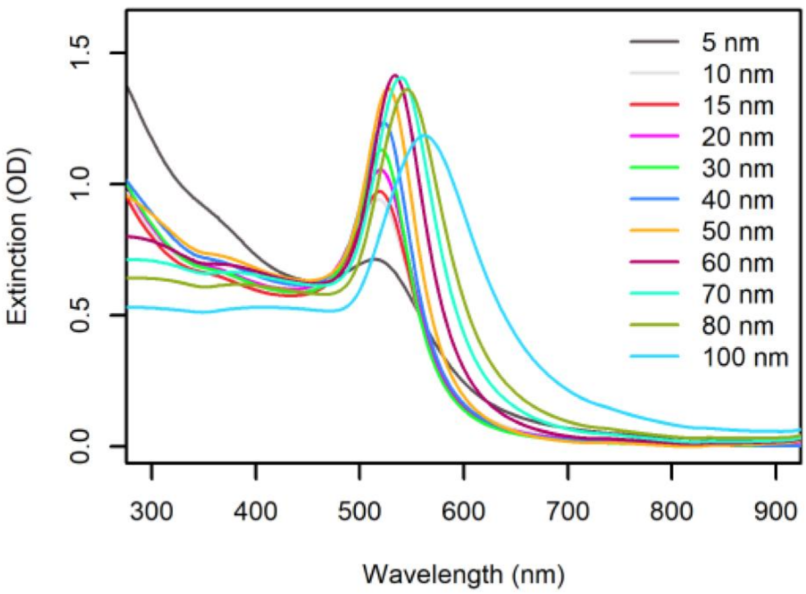

Figure 2.2. Extinction spectra (1 cm pathlength) of commercial 50 µg/mL citrate-capped AuNP colloids (nanoComposix, San Diego, CA, USA) (Ferrari et al., 2023).

nanoparticles also have significantly greater volume, which reduces the absorption cross-section value when it is normalized per particle volume. These to opposing effects help explain the overall size-dependent trend observed in the absorption cross-section per particle volume. For very small nanoparticles, where the particle diameter is significantly smaller than the wavelength of incident electromagnetic wave, only the electric dipole mode can be excited. When the size becomes comparable to wavelength, higher-order multipolar modes emerge and contribute to the plasmonic response (Parisa & Majid, 2021).

These results confirm that particle size is a critical factor that governs the balance between absorption and scattering in plasmonic catalysis. Consequently, controlling size is fundamental in designing and optimizing plasmonic materials for application such as photocatalysist.

2.3. Plasmonic Catalysis

In the recent years, the concept of plasmonic catalysis has gained significant attention as a highly promises strategy to speed up and redirect the pathways of chemical reaction. This approach relies on three fundamental mechanisms: (i) phototermal heating,

(ii) hot carrier injection, (iii) enhancement electromagnetic near-field effect (Kamarudheen et al., 2018).

2.3.1. Phototermal Heating

Metal nanoparticles effeciently convert light into localized heat through the excitation and rapid ultrafast damping of surface plasmons. When the incident light energy matches the LSPR wavelength, a strong resonant occurs, resulting in strong optical absorption and highly enhanced local electromagnetic fields (Brongersma et al., 2015). During this excitation, conduction electron are collectively displaced from their equilibrium positions and oscillate in phase with the external electromagnetic wave, concentrating charge at the nanoparticle surface. The resultant plasmon excitation of the LSPR global non-equilibrium and the induced oscillations rapidly dephase and decay through ultrafast relaxation pathways (**Figure 2.3**). To return the system to equilibrium, the absorbed energy by the electrons can relax through the radiative re-emissions of photon or the non-radiative process such as Landau damping, where the plasmon energy is converted into energetic electron-hole pairs (Zhou et al., 2024).

During the quantum mechanical process of Landau damping, the energy of an excited plasmon is directly transferred to a single electron-hole pair within an ultrafast timescale of approximately 1–100 fs (Brongersma et al., 2015). After photon absorption, the electrons do not immediately reach thermal equilibrium but instead forming an energy distribution, which resembles a double-step function in electron occupancy. Over hundreds of femtoseconds, rapid electron-scattering causes the energy to redistribute among the electrons until they reach a high temperatures Fermi-Dirac distribution. Finally, electron-phonon coupling begins to transfer energy from the hot electrons to the lattice, slowly reducing the electron temperature over a timescale of a few picoseconds (Baffou et al., 2020).

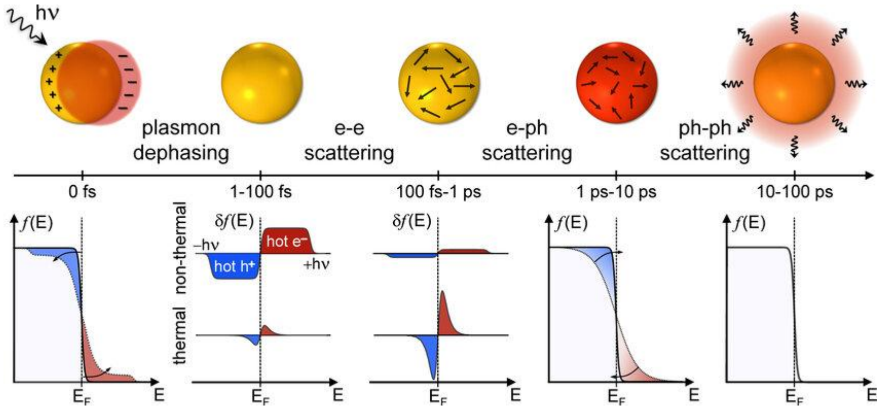

Figure 2.3. Schematic illustration of the relaxation pathways following photoexcitation in a plasmonic nanoparticle (Schirato et al., 2023).

2.3.2. Hot Carrier Injection

Localized surface plasmon resonance (LSPR) oscillate shortly and dephase rapidly into hot charge carriers (electrons and holes). These hot carriers initially display a highly non-thermal energy distribution and then evolves thermally through internal thermalization and subsquent relaxation processes. The generation and relaxation of hot carriers in metal nanoparticles are strongly influenced by quantum and discrete nature of plasmon excitation. Due to the discrete character of plasmon excitation, only a small number of nanoparticles generate hot carriers under CW illumination. In contrast, pulsed laser can excite multiple LSPR on a single nanoparticles, resulting in a significat increase in the amount of generated hot carriers. As shown in **Figure 2.4**, these hot carriers form through interband and intraband electronic transitions and possess sufficient energy to participate directly in chemical reactions (Lei et al., 2024). They possess are extremely short-lived, undergoing ultra-fast thermalization within femtoseconds to picoseconds through scattering processes that turn their excess energy into lattice heat (Kamarudheen et al., 2018).

The ability to efficiently tune and harvest these hot carriers forms the technological basis for numerous advanced applications. Once harvested, these energetic carriers become the key that trigger photocatalytic redox reactions on the nanoparticle surface, allowing light to drive chemical synthesis and environmental remediation under light (Khurgin et al., 2024).

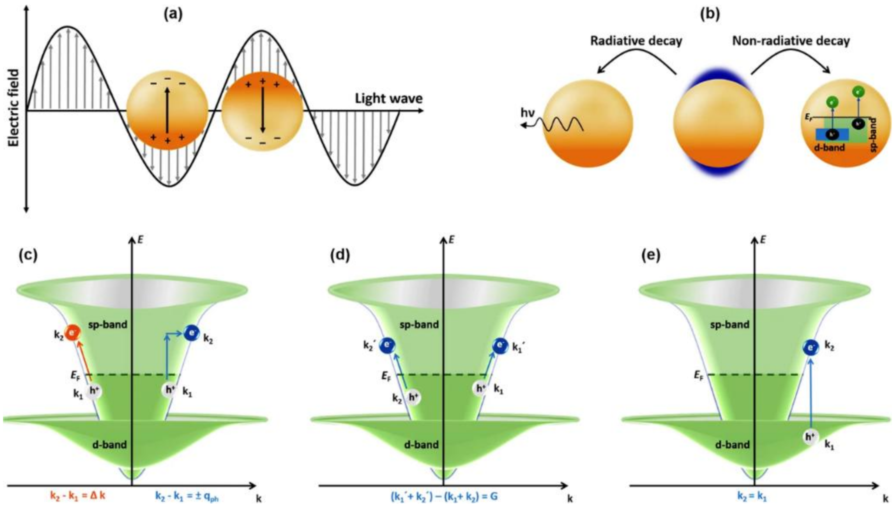

Figure 2.4. a) Interaction of electromagnetic radiation with metal nanoparticle driving a Localized Surface Plasmon Resonance within nanoparticle. (b) Dissipation of strong electric fields near the surface of the metal nanoparticles branches out through either radiative or non-radiative decay. In the framework band transition, several electronics transition is possible: (c) Indirect, intraband s-to-s transitions assisted by phonons (blue arrow) and by surface-collision processes (orange arrow); (d) Indirect, intraband *s*-to-*s* transisitions assisted by electron-electron scattering, and; (e) Direct, interband *d*-to-*s* transition (Ahlawat et al., 2021).

2.3.3. Enhancement Electromagnetic Near-field Effect

The excitation of LSPR in metal nanoparticles generates strongly confined electromagnetic near-fields around the particle surface, forming highly intense optical “hotspots” that concentrate electromagnetic energy at sub-nanometer regions, as illustrated in **Figure 2.5**. This electromagnetic near-field enhancement improves light absorption of the target reactant and promotes molecular electronic excitation without the need for direct charge transfer. As a result, plasmonic near-field enhancement facilitates photochemical reaction pathways by increasing the probability of reactant excitation near the nanoparticle surface by contributing to plasmon-mediated catalytic activity via absoprtion enhancement (Han et al., 2021).

The magnitude and spatial distribution of the near-field response are strongly dependent on the plasmon resonance wavelength, particle shape, and atomic-scale surface structure, which determine the position and intensity of active catalytic sites (Richard-Lacroix & Deckert, 2020).

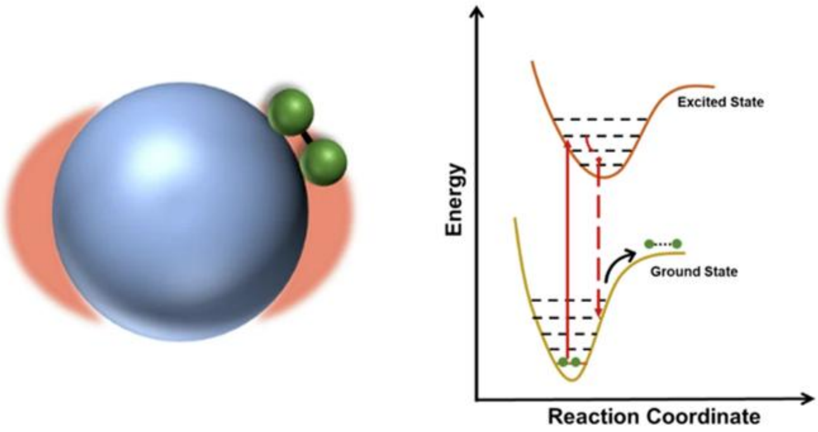

Figure 2.5. Electromagnetic near-field enhancement around a plasmonic nanoparticle increase the probability of molecular excitation via absorption enhancement (Verma et al., 2024).

2.4. Bimetallic Nanoparticles

Bimetallic nanoparticles have attracted huge attention as compared to monometallic nanoparticle in catalysis and plasmonics due to their enhanced performanced and tunable properties. Bimetallic nanoparticles can be synthesized with different size, shape, and structure (Goswami et al., 2025). Traditionally, the optimization of bimetallic nanoparticles consisting of one active and one selective metal has been achieved by varying the metal composition, though this often requires compromising between the catalytic properties of the two metals (van der Hoeven et al., 2021).

While most studies in plasmonic nanomaterials have focused on optically properties monometallic nanoparticles such as gold, silver, or copper, bimetallic nanoparticles offer greater control over plasmonic properties (Srinoi et al., 2018). These nanoparticles are often synthesized as monodisperse spheres with size ranging from a few nanometers to beyond one hundred nanometers in diameter. Because these spherical nanoparticles often display surface plasmon resonance (SPR) response that are limited to a narrow range within the visible spectrum, new geometric shapes including rods, cubics, and core@shell structures have been introduced to expand and enhance their absorption properties (Mcoyi et al., 2024).

Core@shell nanoparticles are often designed with a metal core coated by an additional metal shell layer. These bimetallic nanoparticles show adjustable optic properties because electric coupling between the metals (Li et al., 2021). The optical properties of the core@shell nanoparticles can be tunable by changing the size of core

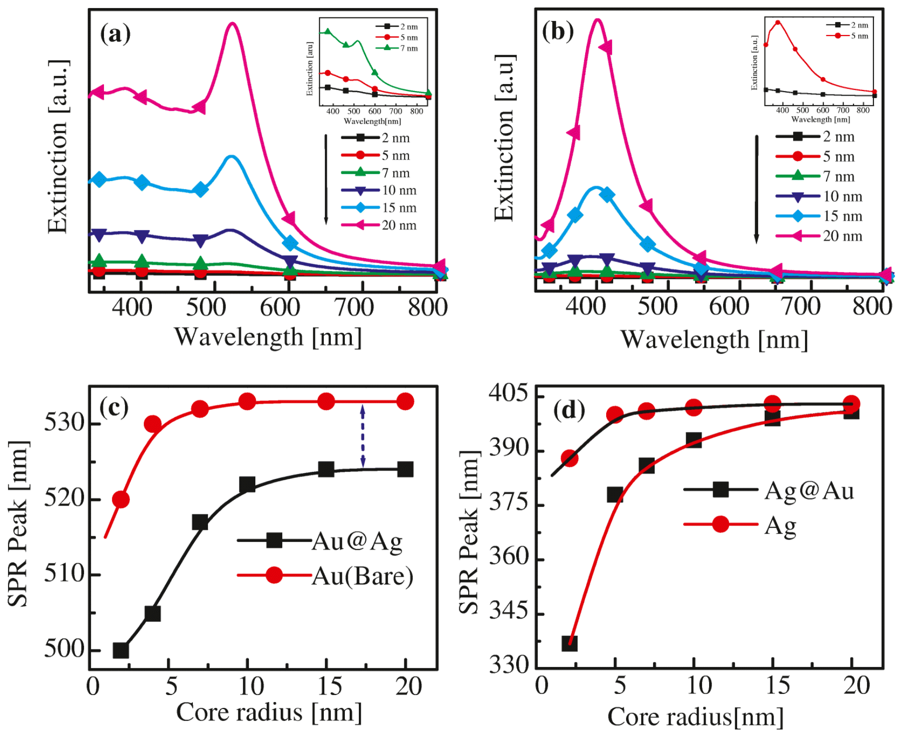

Figure 2.6. The optical extinction spectra for Au@Ag (a) and Ag@Au (b) core@shell nanoparticles with varying core sizes and a fixed 1 nm shell. The insets show few spectra for core sizes of 2, 5, 7 nm in the case of Au@Ag and for 2 and 5 nm core sizes in the case of Ag@Au NPs. (c) and (d) represent how the surface plasmon resonance (SPR) peaks shift as the core radius increases, including a comparison with the peak positions of the corresponding bare nanoparticles (Majhi & Kuiri, 2020).

gold nanoparticles. According to research paper called "Enhancement of spectral shift of plasmon resonance in bimetallic noble metal nanopartilces in core@shell structure" the interaction between gold cores and silver shells inside core@shell nanoparticles dictates their optical behavior, particularly their localized surface plasmon resonance (LSPR). As shown in **Figure 2.6**, this work reveals that varying core radius changing optical extinction spectra with wavelength (Majhi & Kuiri, 2020).

The graph shown that SPR bands in the extinction spectra can be observed as the core radius increase. To further validate the significance of core size, the system was inverted to a silver core with a gold shell while keeping the shell thickness constant at 1 nm, and the same variations in core radius was analyzed. As shown in figure, SPR bands are observed as it is expected. These results proves that larger core sizes enhance the visibility of SPR peaks, indicating that core radius plays a critical role when the shell thickness is held constant, and that the shell thickness only slightly affect the apperance of SPR peaks in the extinction spectra (Majhi & Kuiri, 2020).

2.5. Seed-mediated Synthesis of Gold Nanoparticles

Synthesis of size and shape-controlled gold nanoparticle in principle follows 2 main steps: (i) seed formation via rapid reduction and followed by (ii) particle growth via slow reduction process.

For step (i), gold seeds were prepared by rapidly injecting a freshly prepared $NaBH_4$ solution into an aqueous solution containing $HAuCl_4$ and CTAB. Upon the addition of $NaBH_4$, the solution immediately turned brown, indicating the rapid formation of small gold nuclei due to the reduction of Au^{3+} ions into metallic gold (Au^0). In this process, $NaBH_4$ acted as a strong reducing agent that rapidly generated numerous nuclei within a short period of time, resulting in the formation of small seed particles suitable for subsequent seed-mediated growth. The appearance of a brownish color is commonly associated with the formation of nanosized gold seeds exhibiting localized surface plasmon resonance (LSPR).

In addition, CTAB/CTAC played an important role as a cationic surfactant during seed formation. The CTAB/CTAC molecules adsorbed onto the surface of the newly formed gold seeds, providing colloidal stability through electrostatic repulsion and thereby minimizing nanoparticle aggregation. This surfactant layer was important for preserving nanoparticle dispersity and promoting the formation of relatively uniform seed particles, which subsequently acted as nucleation sites in the following growth stage (Khan et al., 2013).

After the formation of the initial seeds, in the step (ii) the gold nanoparticles were further enlarged through a seed-mediated growth process to obtain intermediate gold nanospheres. During this stage, $HAuCl_4$ functioned as the gold precursor, whereas

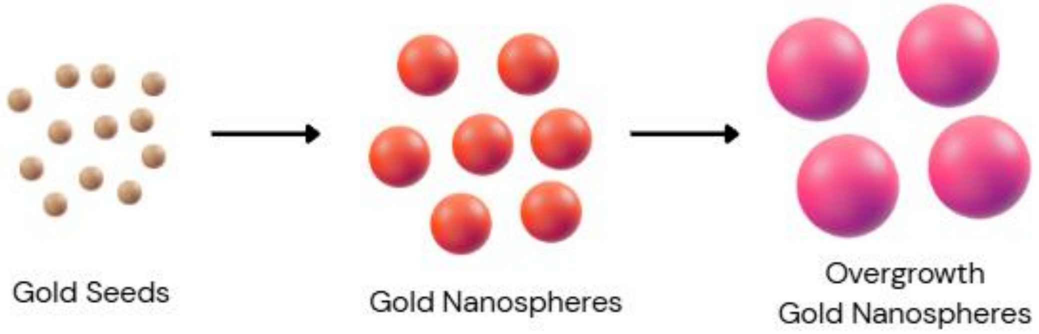

Figure 2.7. The main steps of the synthesis of spherical gold nanoparticles.

ascorbic acid acted as a mild reducing agent that slowly reduced Au^{3+} ions in the presence of the pre-existing gold seeds. Unlike $NaBH_4$, the slower reduction kinetics of ascorbic acid favor heterogeneous growth on the surface of existing nanoparticles rather than rapid secondary nucleation in solution. Consequently, the newly formed gold atoms preferentially deposit onto the surfaces of the pre-existing gold seeds, allowing controlled enlargement of the nanoparticles while preserving their predominantly spherical shape, as shown in **Figure 2.7**. This separation between the nucleation and growth stages is one of the key advantages of seed-mediated synthesis, as it allows improved control over nanoparticle size distribution and morphology (Jana et al., 2001). Moreover, further enlargement is also possible by repeating step (ii).

2.6. Scanning Electron Microscopy

Scanning electron microscopy (SEM) will be employed to investigate the size variation and morphology of the synthesized gold nanoparticles. The SEM analysis provides high-resolution images that enables direct observation of the the particle size, shape, and size distribution. It is expected that variation in synthesis conditions would influence the average particle diameter and size uniformity of the gold nanoparticles. The SEM images of gold nanoparticles with different average diameters are represented in **Figure** , confirming the successful size-controlled synthesis and revealing structural features that may affect the optical properties of the gold nanoparticles.

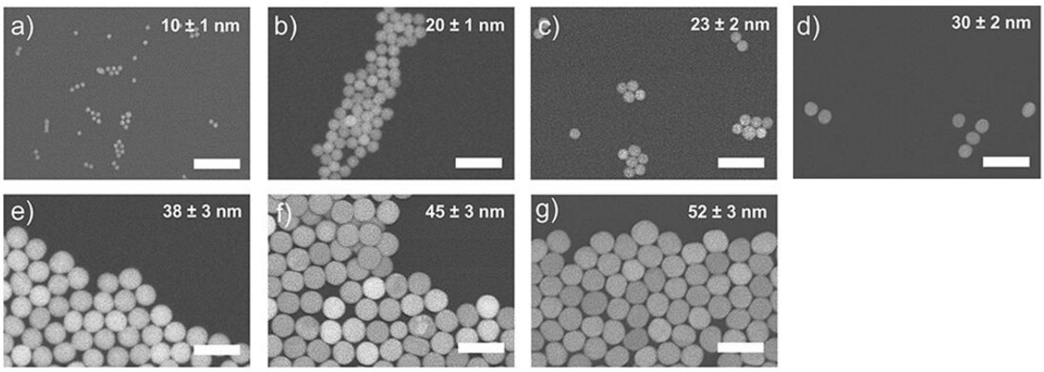

Figure 2.8. Structural and optical characterization of gold nanoparticles with different diameters. Representative SEM images of gold nanoparticles with different average diameter of a) 10 ± 1, (b) 20 ± 1, (c) 23 ± 2, (d) 30 ± 2, (e) 38 ± 3, (f) 45 ± 3, and (g) 52 ± 3 nm (Nguyen & Baldi, 2025b).

Furthermore, SEM is employed to investigate the growth behavior of the silver shell in Au@Ag core@shell nanoparticles. As shown in **Figure** , SEM images of Au@ZnO core@shell nanoparticles are presented as a representative example, illustrating successful shell formation on Au cores, as evidenced by the LSPR redshift and the confirmation of the core@shell structure by SEM analysis. In addition, as shown in **2.10**, SEM images of Au@ZnO core@shell nanoparticles synthesized from gold cores with varying diameters and shapes illustrate how the initial core size influences the final particle size and morphology after shell growth.

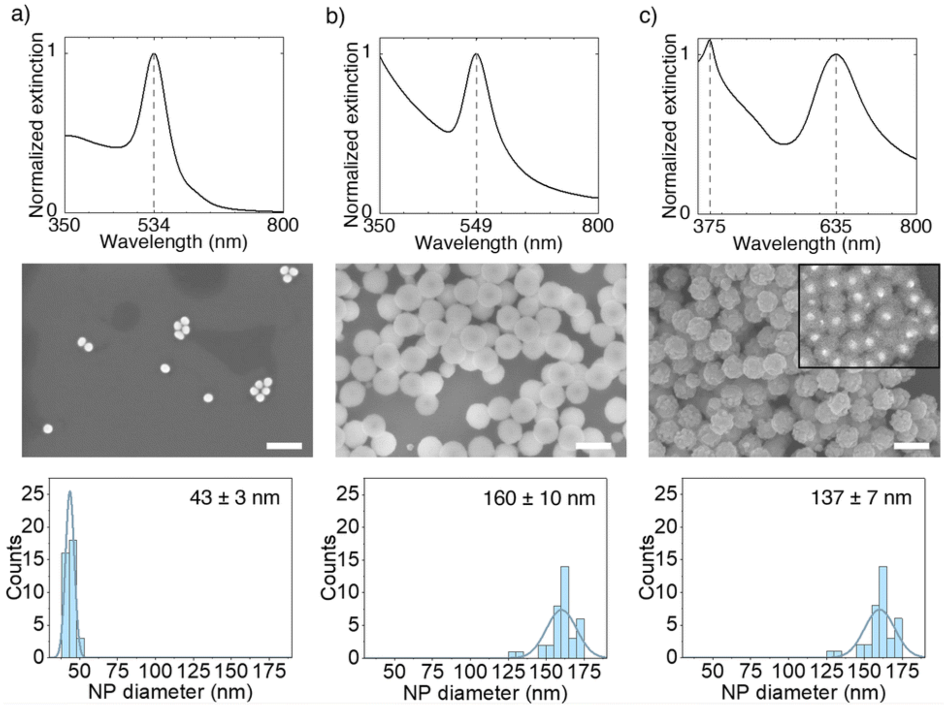

Figure 2.9. Normalized extinction spectrum (top row), secondary electron and backscattered electron SEM images (middle row), and size distributions (bottom row) of (a) as-synthesized AuNPs with an average diameter of 43 nm, (b) Au@Zn(OH)$_2$ nanoparticles after hydrothermal synthesis with an average shell thickness of 59 nm, and (c) Au@ZnO nanoparticles after calcination with an average shell thickness of 47 nm. All scale bars are 200 nm (Nguyen et al., 2025).

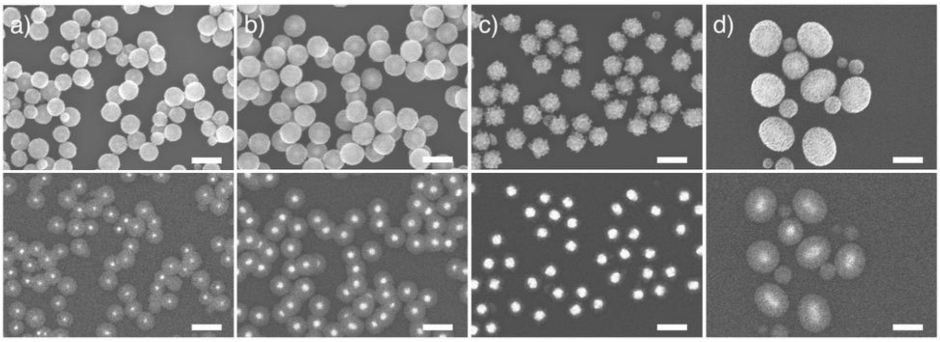

Figure 2.10. Secondary electron (top row) and backscattered electron (bottom row) SEM images of Au@ZnO nanoparticles with spherical cores of (a) 25 nm, (b) 43 nm, and (c) 69 nm, and nanorod cores of (d) 54 × 118 nm. All scale bars are 200 nm (Nguyen et al., 2025).

CHAPTER 3
RESEARCH METHOD

3.1. Research Methodologies

The research methodology that were used in this research is illustrated in **Figure 3.1**. In this research, spherical gold nanoparticles were synthesized indepedently rather than using gold nanoparticles commercial. A bottom-up seed-mediated growth approach wass selected to enable precise control over nanoparticle size and the choice of stabilizing agents, which were crutial parameters for the subsquent-heat/laser-induced core@shell synthesis experiment.

The synthesized gold nanoparticles were used as seeds for gold@silver core@shell nanoparticle formation under under a pulsed laser irradiation at excitation wavelength 532 nm, while varying the laser power from 50 mW to 400 mW. To investigate the effect of laser power, we started with 532 nm-sized nanoparticle, deliberately to match resonance with the laser. The power sweep was also imperative to deduce the power threshold sufficient to induce Ag shell growth. Subsequently, the effect of gold core size variation on the formation of gold@silver core@shell was investigated. Five different gold nanoparticles were used and classified according to their LSPR peak positions relative to 532 nm: (i) two core size with LSPR peaks < 532 nm (520 nm and 525 nm), (ii) one core size with LSPR peak at 532 nm, (iii) two core size with LSPR peaks > 532 nm (536 nm and 540 nm). This effect of core size variation is investigated using a 532 nm pulsed laser at a fixed laser power of 400 mW.

Despite that the actual image (for example by SEM) of the nanoparticles was not provided in this project, I validated the synthesized spherical gold nanoparticle by observing the plasmonics spectra and match them with Mie-scattering simulation for to calculate the diameter.

The shell growth process were monitored in real-time using UV–vis spectroscopy by tracking the evolution and shift of the LSPR peak during the reaction. Finally, the experimental data were processed, visualized, and analyzed to evaluate the effects of heating mode, laser irradiation, and core size variation on the gold@silver core@shell growth behavior.

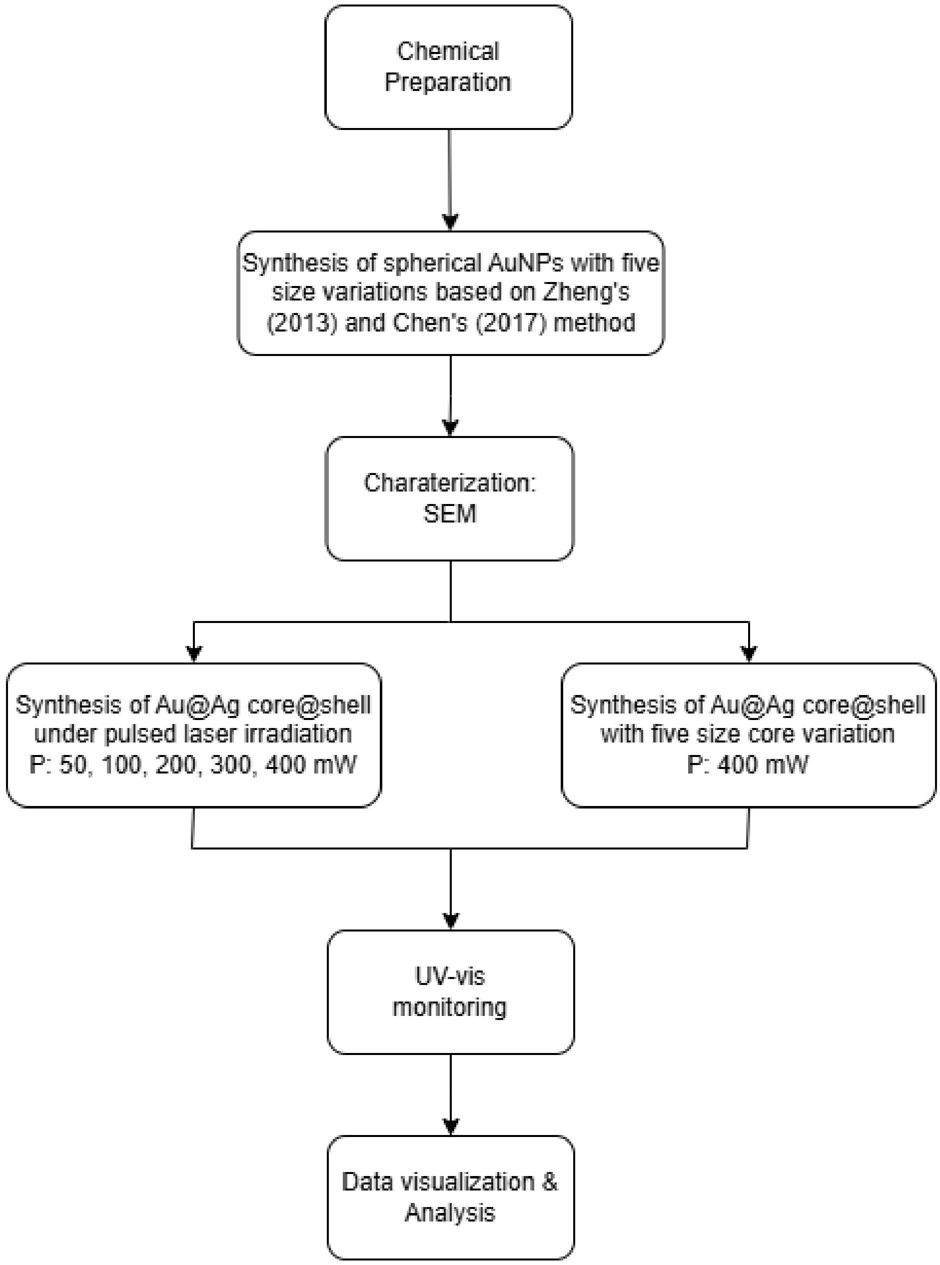

Figure 3.1. The schematic diagram of the research methodologies in the present work.

3.2. Experimental Procedures

The experimental procedures that are conducted are divided into three main steps.

3.3. Synthesis of Spherical Gold Nanoparticles

One of the challenge in this project is to synthesize gold nanoparticle as core with size control and excellent reproducibility. In the first attempt, I adapted method reported by Zheng et al., 2013. With this method, however, I have to rely on extra infrastructure i.e. syringe pump that also limits the scalability. After many attempts, I concluded that this method suffered a low reproducibility. Therefore, later I took initiative to explore method reported Chen et al., 2017. Nevertheless, in this section I discussed the steps from both methods.

For Zheng et al., 2013 method, the synthesis procedure is performed in three steps, as followed.

The *first* step was to rapidly add a fresh $NaBH_4$ (10 mM, 0.6 mL) solution into a mixed solution containing $HAuCl_4$ (0.25 mM) and CTAB (100 mM), which immediately produced a brown-color solution. The solution was then stirred at 300 rpm for 2 minutes and then left undisturbed at room temperature (27 ° C) for 3 hours to ensure full decomposition of the remaining $NaBH_4$.

The *second* step was preparation of a solution containing CTAC (200 mM, 2 mL), ascorbic acid (100 mM, 1.5 mL), and the CTAB-capped gold clusters in glass vial, followed by one-shot injection of $HAuCl_4$ solution (0.5 mM). The mixture was kept at room temperature (27 °C) for 15 minutes. The product was obtained after centrifugation at 14500 rpm for 30 minutes, and the resulting gold nanospheres were redispersed in CTAC solution (20 mM, 1 mL) to use as seeds for second round of growth.

The *third* and final step involved mixing CTAC (100 mM, 1 mL), ascorbic acid (10 mM, 130 μL), and the previously prepared seeds in glass vial. In this step, $HAuCl_4$ is added dropwise at 2 mL h^{-1} using a syringe pump (MedimiG OSP-500) to control the growth rate of nanoparticles and avoid non-isotropic morphological changes. After the addition, the mixture was allowed to proceed at 27 °C for 10 minutes. The resulting gold nanospheres was collected by centifugation at 12000 rpm for 10 minutes and redispered in CTAC solution (10 mM, 3 mL).

In seed-mediated methods, pre-synthesized gold nanoparticle seeds were used to grow larger nanoparticles thorugh the addition of golds ions. As shown in **Table 3.1**, the size of the nanoparticles can be controlled by varying the concentration of the gold ions.

Table 3.1. Summary of synthesis parameters and nanoparticle size variation of gold nanopsheres adopted from Zheng et al., 2013.

d_{av} [nm]	σ [%]	d_{hydro} [nm]	σ_{hydro} [%]	Type of seed	V [[μL]
5.2	0.52	n/a	n/a	Cluster	1000

8.0	0.54	n/a	n/a		500
10	0.81	n/a	n/a		50
16	1.6	26.81	35.6		20
15	1.1	26.25	42.2	10-nm sphere	300
23	1.1	36.46	34.8		100
46	1.9	58.55	20.2		10
80	1.3	103.0	26.8		5
70	1.5	103.5	14.1	46-nm sphere	1500
100	3.1	131.9	23.2		500
150	4.8	198.0	29.2		50

Due to the limited reproducibility observed during preliminary experiments, an alternative growth strategy adapted from Chen et al., 2017 was subsequently employed in this study.

In this modified growth procedure, the gold growth solution was prepared by mixing CTAC (0.2 M, 5 mL), $HAuCl_4$ (0.01 M, 0.25 mL), and deionized water (4.65 mL) in a glass vial. Subsequently, freshly prepared ascorbic acid (0.04 M, 100 μL) solution was added under vertical agitation until the solution became colorless, indicating the reduction of Au^{3+} ions into Au^{+} species. Previously pre-synthesized gold nanoparticles seeds were then introduced into the prepared growth solution under agitation and allowed to age for 30 min without disturbance. Gold nanoparticles with different LSPR peaks were prepared by varying the amount of gold seed solution introduced into the growth system.

3.4. Synthesis of Core@shell Gold@silver Nanoparticles

The core@shell gold@silver nanoparticles synthes as model catalysis is adopted from the method reported by Nguyen and Baldi (2025).

The *first* step was to add the previously prepared spherical gold nanoparticles solution (200 μL) into CTAC solution (0.1 mM, 2 mL) under stirring at 500 rpm. Subsequently, BSPP solution (375 mM, 10 μL), $AgNO_3$ solution (150 mM, 10 μL), and ascorbic acid solution (150 mM, 31 μL) were added with 3 minutes interval between each addition while maintaining stirring.

The *second* step was to trigger core@shell formation either using conventional heating or using nanosecond pulsed laser irradiation.

In synthesis under nanosecond pulsed laser, the previously prepared growth solution was transferred into a quartz cuvette ($10 \times 10 \times 40$ mm^2) and placed the cuvette in a cuvette holder for irradiation using a nanosecond pulsed Nd:YAG laser (repetition rate = 100 Hz; Litron Lasers) that operates at excitation wavelength 532 nm and output powers of 50, 100, 200, 300, 400 mW. To avoid the laser beam being reflected back to the laser, the laser beam path is deflected using a mirror. Throughout the irradiation process, the absorbance spectrum was measured with halogen light source (SLS210L/M, Thorlabs) and spectrometer (Aurora 4000, CNI Laser).

The size variation investigated in this study was consisted of five different particle size, as summarized in **Table 3.2**.

Table 3.2. Summary of gold nanoparticle core size variations used in this study

Sample	LSPR Peak (nm)
1	520 nm
2	525 nm
3	532 nm
4	536 nm
5	540 nm

.

3.5. Kinetics Study using Real-time UV-Vis Absorbance Spectroscopy

The kinetics study was performed using real-time UV-Vis spectroscopy (AvaSpec-ULS2048CL-EVO, Avantes). This experiment was carried out using a home-made cuvette holder with a PID temperature control and magnetic stirrer (**Figure 3.2**). The UV-Vis transmission spectra was used to monitor the LSPR peak during the in-situ core@shell synthesis using PySpecTrace, a Python-based GUI developed by (Anam, 2026) to quantify the silver shell growth rate from the observed LSPR peak shifts and then compare the kinetics between five different gold core sizes under pulsed laser irradiation.

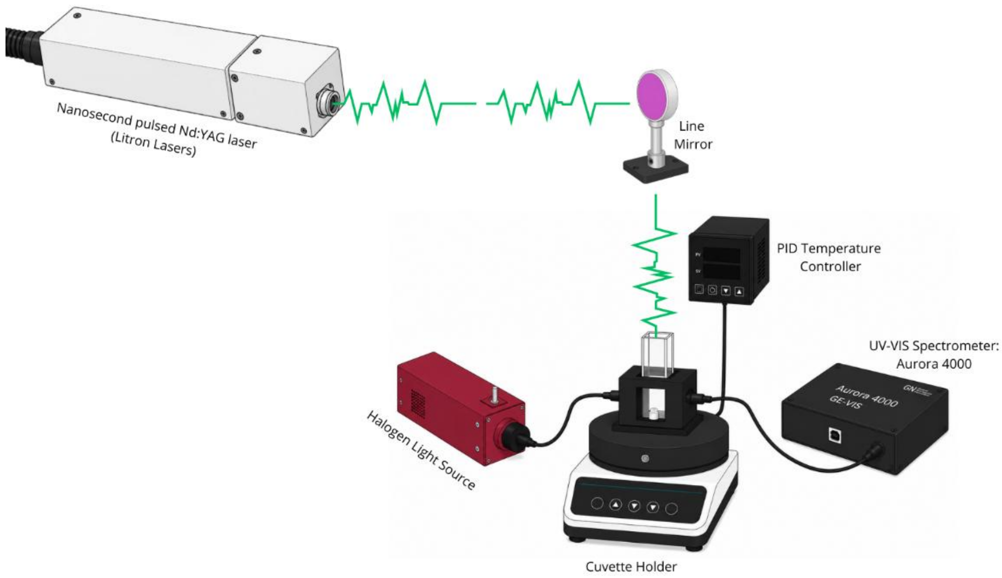

Figure 3.2. Schematic illustration of home-made cuvette holder for real-time UV-Vis kinetic measurements.

CHAPTER 4
RESULT AND DISCUSSION

4.1. Synthesis of Spherical Gold Nanoparticles

Spherical gold nanoparticles with LSPR peak around 532 nm were succesfully synthesized using a seed-mediated approach adapted from previously work (Zheng et al., 2013).

The successful growth of gold nanospheres was indicated by the gradual color transformation of the solution from brownish-red to wine-red after the addition of the gold seed solution into the growth medium. This color evolution suggests an increase in nanoparticle size and the corresponding enhancement of LSPR within the visible region, as shown in **Figure 4.1**. The resulting gold nanospheres exhibited an LSPR peak around 517 nm and were subsequently utilized as intermediate seeds for the secondary overgrowth process in the following synthesis stage. In this method, controlled nanoparticle overgrowth was achieved by gradually introducing $HAuCl_4$ into a growth solution containing CTAC, ascorbic acid, and pre- formed gold nanospheres using a syringe pump. The slow injection rate of the gold precursor allowed more controlled

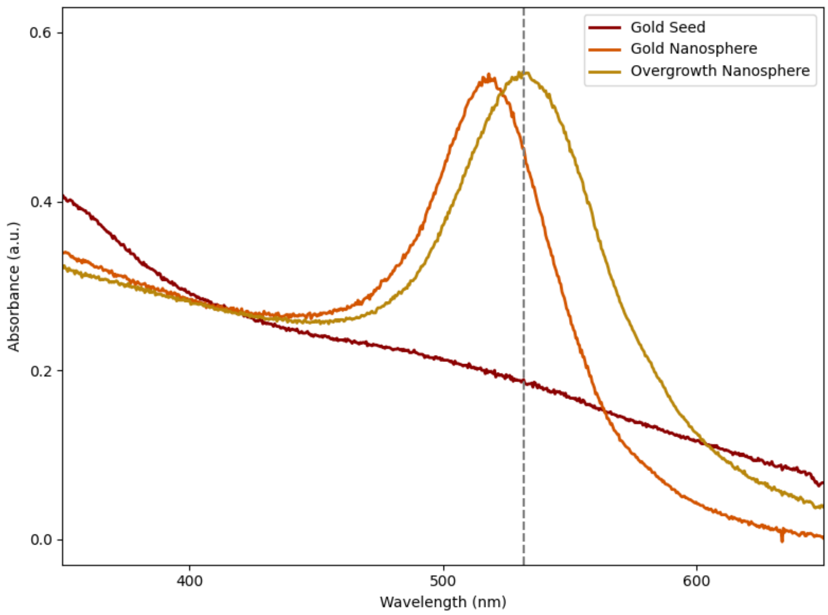

Figure 4.1. The absorbance spectrum of gold seeds, intermediate gold nanospheres, and overgrown gold nanospheres. The gradual red-shift of the LSPR peak from the intermediate growth stage with LSPR peak around 517 nm to the overgrowth stage with LSPR peak around 532 nm.

deposition of gold atoms onto the surfaces of the existing nanoparticles, thereby minimizing excessive secondary nucleation and facilitating the growth of particles with a predominantly spherical morphology. Using this approach, gold nanospheres exhibiting an LSPR peak around 532 nm were successfully synthesized, as shown in **Figure 4.1**.

Although the method successfully produced larger gold nanoparticles, the resulting colloidal solutions exhibited limited reproducibility between synthesis batches. Variations in optical response and particle growth behavior were observed during repeated experiments, indicating reduced consistency of the overgrowth process under the employed experimental conditions, as shown in **Figure 4.2a**.

Therefore, an alternative overgrowth strategy adapted from Chen et al., 2017 was subsequently implemented to enhance nanoparticle growth consistency and improve synthesis reproducibility. Unlike the previous overgrowth method, this modified approach employed pre-formed gold nanospheres as intermediate seeds within a comparatively simpler and more controlled growth environment consisting of CTAC, $HAuCl_4$, and ascorbic acid. This methode approach did not rely on continuous precursor injection using a syringe pump during nanoparticle growth. Instead, nanoparticle enlargement occurred under more homogeneous reaction conditions through the gradual reduction of Au^{3+} ions by ascorbic acid, followed by continuous gold deposition onto the seed surface.

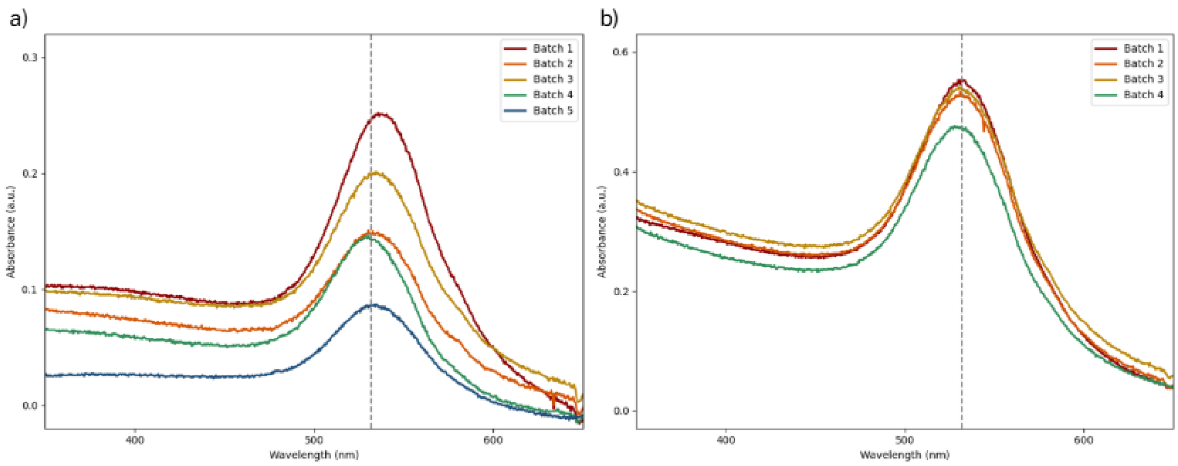

Figure 4.2. The absorbance spectrum of gold nanoparticles synthesized using the overgrowth method adapted from (a) Zheng (2013) and (b) Chen (2017). Method (a) yield a less reproducible end-products than that of method (b).

A shown in **Figure 4.2b**, the absence of continuous precursor injection reduced the sensitivity of the growth process toward minor experimental fluctuations, thereby enabling more stable nanoparticle enlargement, improved batch-to-batch consistency, and better synthesis reproducibility. Using this modified growth strategy, five gold nanoparticle samples with LSPR peaks around 520 nm, 525 nm, 532 nm, 536 nm, and 540 nm were successfully synthesized, as shown in **Figure 4.3**. The variation in LSPR peak positions indicates differences in nanoparticle size and optical response resulting from the controlled overgrowth process. In general, the observed red-shift toward longer wavelengths suggests an increase in nanoparticle size due to the continued deposition of gold atoms onto the existing nanoparticle surface.

Despite succesful peak position control using Chen's method, there is caveat worth to mention here. The 540 nm sample shows significant asymmetrical peak broadening that indicates non-isotropic transformation from spherical to non-spherical nanoparticle shape. This highlight limitation of Chen's method at long LSPR wavelength. This limitation stems from Chen's rapid injection method that is less controlable, in contrast to Zheng's gradual injection.

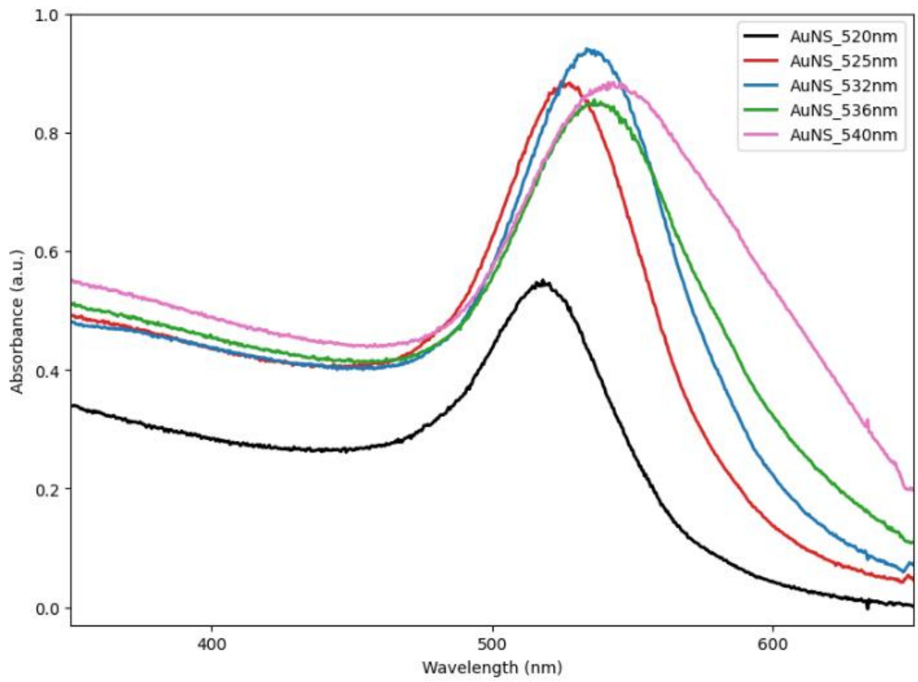

Figure 4.3. The absorbance spectrum of gold nanoparticles synthesized using the modified overgrowth strategy, exhibiting LSPR peaks around 520 nm, 525 nm, 532 nm, 536 nm, and 540 nm.

4.2. Validation of LSPR Peaks Using Mie Theory

Numerical calculations based on Mie's theory method were employed to provide further insight into the size-dependent optical behavior of the synthesized gold nanoparticles and to predict the optical evolution during the gold@silver core@shell growth process.

The Mie theory calculations for spherical gold nanoparticles and gold@silver core@shell structures were performed using MATLAB scripts adapted from Baldi (2022, 2024) following the theoretical formalism developed by Bohren and Huffman (1998). The calculations and simulations were conducted using a surrounding refractive index of 1.333 to represent water, while the dielectric functions of gold and silver were obtained from the optical constants reported by Johnson & Christy.

In the Mie theory calculations for spherical gold nanoparticles, the particle radius was systematically varied to evaluate the relationship between nanoparticle size and the corresponding LSPR peak position, as summarized in **Table 4.1**.

Table 4.1. LSPR peaks and corresponding gold nanoparticle diameters based on Mie Theory.

LSPR Peak (nm)	Particle Diameter (nm)
522	14
525	19
532	27
536	30
540	33

Overall, the calculated results show that increasing the nanoparticle radius causes the LSPR peak to shift toward longer wavelengths while simultaneously increasing the extinction cross-section intensity, as shown in **Figure 4.4**. This behavior is consistent with the plasmonic characteristics of spherical gold nanoparticles, where larger particle sizes enhance light absorption and scattering processes, resulting in stronger plasmon resonance within the visible region.

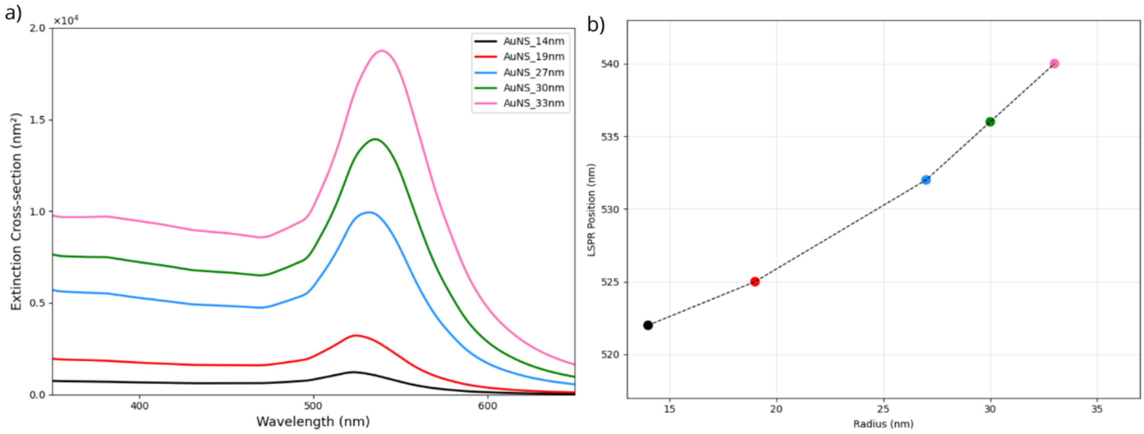

Figure 4.4. Mie theory calculations of spherical gold nanoparticles with different particle radius (~14 to 34 nm) values: (a) calculated extinction spectra and (b) correlation between particle radius and corresponding LSPR peak position.

However, the Mie theory calculations also revealed that gold nanoparticles with a radius smaller than approximately 14 nm exhibited relatively limited LSPR peak variation, remaining peak around 522 nm despite further decreases in particle size. This behavior suggests that, at smaller nanoparticle dimensions, the plasmonic response becomes less sensitive to size reduction within the investigated range. In small spherical gold nanoparticles, the optical response is predominantly governed by absorption rather than scattering effects, resulting in relatively minor spectral shifts for small changes in particle size. Consequently, nanoparticles with a radius below 14 nm tend to exhibit similar LSPR peak positions within the visible region.

This observation also indicates a limitation in estimating very small nanoparticle sizes solely based on LSPR peak positions, since nanoparticles with different small particle radius values may exhibit nearly identical optical responses. Therefore, additional structural characterization techniques such as SEM or TEM are generally required to accurately distinguish nanoparticle sizes within the small-particle regime.

Mie theory calculations were further performed to investigate the optical evolution of gold@silver core@shell nanoparticles during silver shell growth. In this calculation, spherical gold nanoparticles with a radius of approximately 27 nm, corresponding to an LSPR peak around 532 nm, were used as the core material. The silver

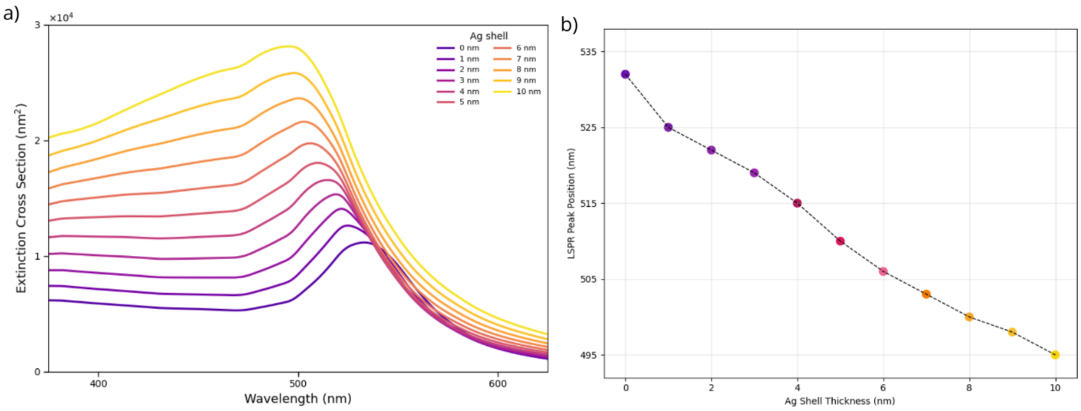

Figure 4.5. Mie theory calculations of gold@silver core@shell nanoparticles using a spherical gold core with a radius of approximately 27 nm: (a) calculated extinction spectra for different silver shell thicknesses and (b) correlation between silver shell thickness and the corresponding LSPR peak position.

shell thickness was systematically varied from 0 nm to 10 nm to evaluate the effect of shell growth on the optical response of the gold@silver core@shell structure.

As shown in **Figure 4.5**, increasing silver shell thickness resulted in a gradual blue-shift of the LSPR peak toward shorter wavelengths along with an increase in extinction cross-section intensity. The calculated spectra indicate that the optical response of the nanoparticles became increasingly dominated by the plasmonic characteristics of silver as the shell thickness increased. This behavior is attributed to the stronger plasmon resonance of silver at shorter wavelengths compared to gold.

The calculations demonstrate that increasing shell thickness progressively shifted the LSPR peak from around 532 nm for gold nanoparticles to around 495 nm for gold@silver core@shell nanoparticles with a 10 nm silver shell. The observed blue-shift confirms the successful modification of the nanoparticle optical properties during shell growth and indicates the increasing contribution of the silver shell to the overall plasmonic response of the core@shell structure.

4.3. Effect of Laser Power on Gold@Silver Core@Shell Synthesis under Nanosecond Pulsed Laser Irradiation

In this study, the effect of nanosecond Nd:YAG pulsed laser irradiation on the plasmon-assisted synthesis of gold@silver core@shell nanoparticles was investigated. The pulsed laser source operated at 532 nm with a beam diameter of approximately 5 mm, a repetition rate of 100 Hz, a vertical polarization, and a pulse duration of approximately 7.1 ns. To investigate the effect of laser intensity on silver shell growth kinetics, the laser power was varied at 50, 100, 200, 300, and 400 mW.

The synthesis process was first carried out by preparing a reaction solution containing gold nanoparticles, silver precursor solution, stabilizing agent, and reducing agent inside a quartz cuvette under continuous magnetic stirring. PID temperature controller was employed to maintain the reaction temperature at room temperature (27 °C) throughout the experiment. Subsequently, the reaction mixture was irradiated using a 532 nm nanosecond pulsed laser for 30 minutes at a laser power of 400 mW, while the optical evolution of the nanoparticles was continuously monitored using in-situ UV-Vis spectroscopy (see **Figure 3.2** for the experimental setup schematic).

During laser irradiation, plasmonic excitation on the surface of the gold nanoparticles generated localized electromagnetic fields and photothermal effects that facilitated the reduction of silver ions and promoted silver deposition onto the nanoparticle surface, resulting in the gradual formation of a silver shell layer surrounding the gold core. As shown in **Figure 4.6**, the reaction solution exhibited a gradual increase in absorbance accompanied by a color transformation toward a yellowish appearance, indicating the progressive growth of the silver shell layer as corroborated by the Mie calculation above.

Following the successful formation of the gold@silver core@shell nanoparticles under 400 mW laser irradiation, the influence of laser power on the shell growth kinetics was subsequently investigated by varying the pulsed laser power to 50, 100, 200, 300, and 400 mW under otherwise identical experimental conditions.

However, direct monitoring of the LSPR peak evolution near 532 nm could not be accurately performed due to the use of a notch filter, which suppressed the spectral region surrounding the excitation wavelength in order to prevent scattered laser light from reaching the spectrometer. Although the spectral region near 532 nm was partially suppressed by the notch filter, the obtained extinction spectra are still presented to qualitatively demonstrate the optical evolution occurring during the shell growth process, as shown in **Figure 4.7**. Therefore, the shell growth process was evaluated through the

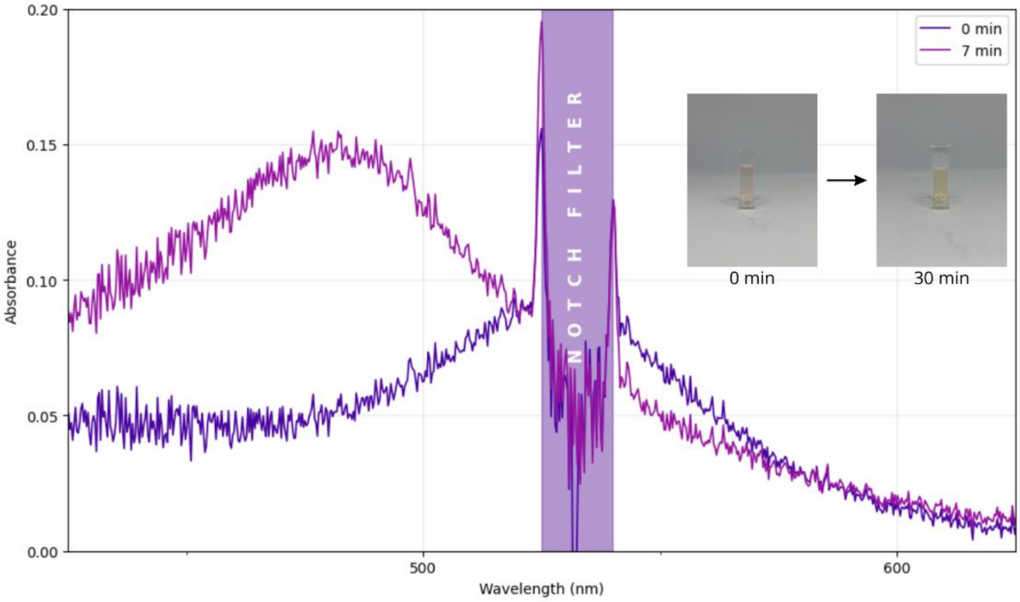

Figure 4.6. The absorbance spectrum of the gold@silver core@shell synthesis under a 532 nm nanosecond pulsed laser irradiation at the power level of 400 mW.

evolution of extinction intensity within the wavelength region of approximately 430–470 nm instead of direct peak tracking (Nguyen & Baldi, 2025).

This spectral region was selected because it is sufficiently blue-shifted from the original gold nanoparticle LSPR while remaining highly sensitive to the formation and growth of the silver shell. As the silver shell gradually formed on the surface of the gold nanoparticles, the extinction contribution within the 430–470 nm region continuously increased. The increase in extinction intensity within this shorter wavelength region indirectly indicates the occurrence of LSPR blue-shift during shell growth, consistent with the Mie theory calculations (**Figure 4.5)**, where increasing silver shell thickness progressively shifted the LSPR peak toward shorter wavelengths.

To quantitatively investigate the shell growth kinetics, the optical evolution within the selected wavelength region was analyzed using extinction contrast (EC), which was calculated as follows:

$$\mathrm{EC(t)} = \frac{\int_{470}^{430} \mathrm{Ext(t,\lambda)\,d\lambda} - \int_{470}^{430} \mathrm{Ext(0,\lambda)\,d\lambda}}{\int_{470}^{430} \mathrm{Ext(0,\lambda)\,d\lambda}} \tag{5}$$

where $\mathrm{Ext(t,\lambda)}$ corresponds to the extinction intensity at irradiation time t and wavelength λ, while $\mathrm{Ext(0,\lambda)}$ corresponds to the initial extinction spectrum before laser

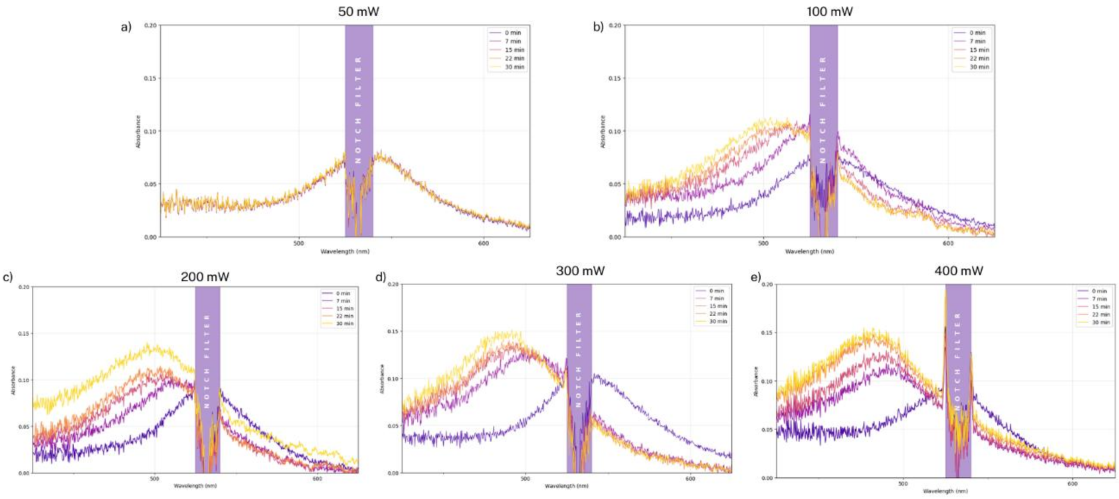

Figure 4.7. Time evolution of the absorbance spectrum of gold@silver core@shell synthesis under nanosecond pulsed laser irradiation at various power levels: (a) 50 mW; (b) 100 mW; (c) 200 mW, (d) 300 mW; (e) 400 mW.

irradiation. The EC parameter therefore describes the relative change in extinction intensity within the selected wavelength region during the shell growth process.

As shown in **Figure 4.8a**, the 50 mW irradiation showed only negligible EC evolution throughout the irradiation period, suggesting that the employed laser intensity was insufficient to significantly promote silver shell growth. However, when the laser power is increased to 100 mW, substantial EC growth were observed, indicating that the power treshold to initiate reaction is 100 mW.

From this threshold, higher laser power produced larger EC enhancement during the irradiation process. The observed increase in EC indicates a stronger extinction contribution from the growing silver shell, as the gradual formation and increase in silver shell thickness progressively shifted the plasmonic response toward shorter wavelengths, resulting in enhanced extinction contribution within the monitored spectral region.

Furthermore, the shell growth kinetics were analyzed using the apparent growth rate constant (k_{app}) obtained by fitting the EC evolution based on the equation:

$$\mathrm{EC(t)} = \mathrm{constant} * \left(1 - \exp\left(-k_{app}\mathrm{t}\right)\right) \quad (6)$$

where t corresponds for irradiation time in seconds.

The k_{app} results show an increasing trend with increasing pulsed laser power, indicating faster silver shell growth kinetics under stronger laser irradiation, as shown in **Figure 4.8b**.

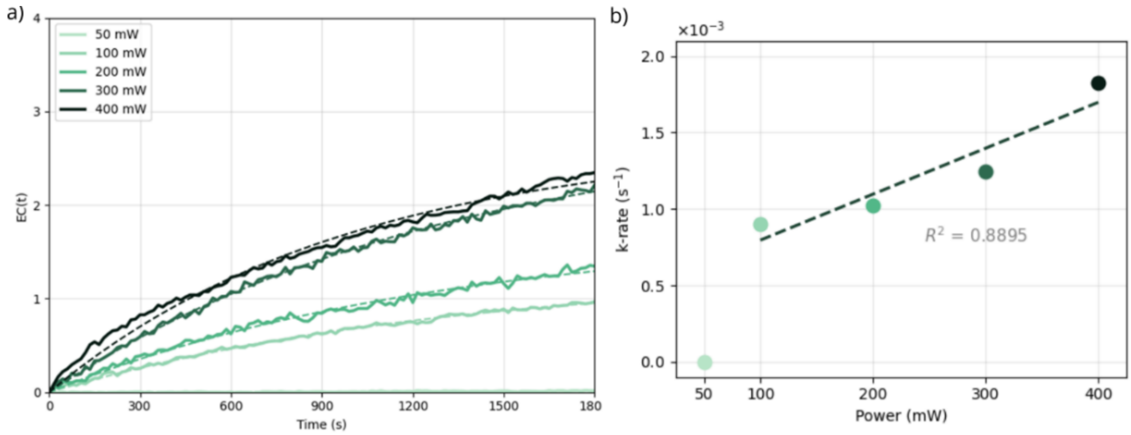

Figure 4.8. (a) Extinction contrast (EC) temporal evolution of gold@silver core@shell nanoparticle growth under different 532 nm nanosecond pulsed laser power irradiation and (b) apparent rate of silver shell growth as a function of laser power.

The increase in k_{app} with laser power is attributed to enhanced plasmonic excitation generated under higher pulsed laser intensity. Stronger laser irradiation can generate more intense localized electromagnetic fields, enhanced light absorption, increased hot-carrier generation, and more significant localized photothermal effects on the surface of the gold nanoparticles. These plasmon-induced effects facilitate faster reduction of silver ions and promote more rapid silver deposition onto the nanoparticle surface, thereby accelerating the shell growth process.

4.4. Effect of Gold Core Size on Gold@Silver Core@Shell Synthesis under Nanosecond Pulsed Laser Irradiation

Following the investigation of laser power dependence, the effect of gold core variation on the gold@silver core@shell growth process was systematically investigated under a constant pulsed laser power of 400 mW. This experiment represents the main focus of the present study, aiming to evaluate how the optical properties of plasmonic gold cores influence the kinetics of silver shell growth under 532 nm nanosecond pulsed laser irradiation. In this experiment, spherical gold nanoparticles exhibiting LSPR peaks around 520 nm, 525 nm, 532 nm, 536 nm, and 540 nm were employed as plasmonic cores to investigate the relationship between the gold core optical response and the silver shell growth behavior. All samples are set to optical density (OD) 0.1 for fair comparison and to limit collective photothermal effects.

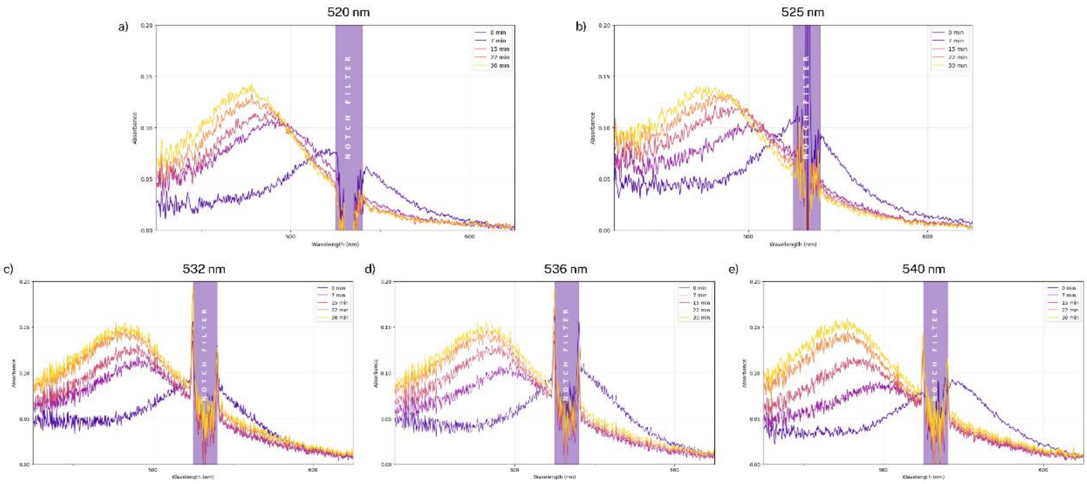

Figure 4.9. Time evolution of the absorbance spectrum of gold@silver core@shell synthesis under nanosecond pulsed laser irradiation at various core sizes: (a) 520 nm; (b) 525 nm; (c) 532 nm, (d) 536 nm; (e) 540 nm.

Similar to the previous experiment, the optical evolution during shell growth was monitored through in-situ UV-Vis spectroscopy. The obtained extinction spectra for different gold core variations are shown in **Figure 4.9**. The shell growth process was analyzed using extinction contrast (EC) within the wavelength region of approximately 430–470 nm, which is highly sensitive to the optical contribution of the growing silver shell.

As shown in **Figure 4.10a**, all gold core variations exhibited gradual EC enhancement with increasing irradiation time, indicating the progressive formation and growth of the silver shell layer during pulsed laser irradiation. Among the investigated samples, gold nanoparticles with longer initial LSPR wavelengths exhibited faster EC evolution. The observed increase in EC indicates stronger extinction contribution from the growing silver shell as the shell thickness progressively increased during the synthesis process. As the silver shell gradually formed on the surface of the gold nanoparticles, the plasmonic response continuously shifted toward shorter wavelengths, resulting in enhanced extinction contribution within the monitored spectral region.

To further quantify the shell growth behavior, the EC evolution curves were fitted using the same exponential growth model described in the previous section to obtain the apparent growth rate constant (k_{app}). The obtained k_{app} values exhibited an increasing trend with increasing initial gold core LSPR wavelength, as shown in **Figure 4.10b**. The

540 nm gold core exhibited the highest apparent growth rate constant, followed sequentially by 536, 532, 525, and 520 nm.

The observed trend of increasing silver shell growth rate (quantified by the extinction contrast, EC) with larger gold nanoparticle diameter under identical pulsed laser irradiation (OD = 0.1 at 532 nm for all samples) can be understood within the framework established by Nguyen & Baldi (2025). In their report, they argue that the predominant mechanism for Ag shell growth is d-band holes generated in the gold core.

In their work, the EC rate tends to decrease for larger gold cores when the nanoparticles are spherical — which contrasts with what we have observed here. However, they also pointed out that faceted nanoparticle shapes can reverse this trend because hot carriers tend to accumulate at sharp surfaces and corners.

This may explain our results. When we reviewed the synthesized nanoparticles, it appears that larger nanoparticles tend to show non-spherical shapes. Further characterization using Scanning Electron Microscopy is required for confirmation, but the current observations strongly indicate that larger nanoparticles tend to adopt non-spherical shapes.

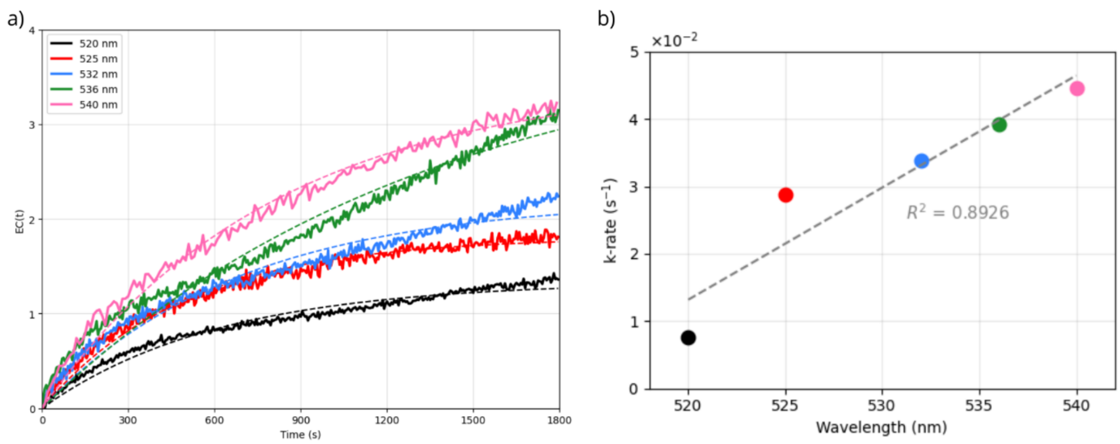

Figure 4.10. (a) Extinction contrast (EC) evolution curves of gold@silver core@shell growth using gold nanoparticle cores with different initial LSPR wavelengths under 532 nm nanosecond pulsed laser irradiation at a constant laser power of 400 mW and (b) apparent rate of silver shell growth as a function of initial LSPR wavelengths.

CHAPTER 5
CONCLUSIONS AND OUTLOOKS

In summary, this study successfully demonstrates the influence of gold nanoparticle core size on pulsed laser-induced plasmonic catalysis during gold@silver core@shell nanoparticle synthesis. The project began with the optimization of spherical gold nanoparticle synthesis using a seed-mediated growth method. Initially, the overgrowth strategy adapted from Zheng et al., (2013) successfully produced larger gold nanoparticles with LSPR peaks around 532 nm. However, the method exhibited limited reproducibility due to the dependence on continuous precursor injection using a syringe pump. Therefore, an alternative growth strategy adapted from Chen et al., (2017) was subsequently implemented, enabling more stable nanoparticle enlargement and improved batch-to-batch reproducibility. Using this modified approach, five gold nanoparticle samples with LSPR peaks around 520 nm, 525 nm, 532 nm, 536 nm, and 540 nm were successfully synthesized (with caveat that the shape may not be perfectly spherical due to inherent nucleation kinetic limitation).. More specifically, the present study yields the following conclusions:

• The optical response of spherical gold nanoparticles strongly depends on nanoparticle size. Increasing nanoparticle size produced a gradual red-shift in the LSPR peak together with enhanced extinction intensity.

• Nanosecond pulsed laser irradiation at 532 nm successfully induced gold@silver core@shell formation, as evidenced by the gradual increase in extinction intensity, the evolution of the extinction contrast (EC), and the visible color transformation of the colloidal solution during irradiation. The shell growth process was associated with a gradual blue-shift of the plasmonic response toward shorter wavelengths due to the increasing contribution of silver plasmon resonance as the shell thickness increased.

• The shell growth kinetics strongly depended on pulsed laser power. Increasing laser power from 50 mW to 400 mW progressively accelerated the shell growth process, as demonstrated by the increase in the apparent growth rate constant. The enhanced shell growth under stronger laser irradiation is attributed to more intense plasmonic excitation, resulting in stronger localized electromagnetic fields, enhanced light absorption, increased hot-carrier generation, and more significant localized photothermal effects that promote faster reduction of silver ions and more rapid silver deposition onto the nanoparticle surface.

• The effect of gold core size on shell growth kinetics was successfully demonstrated under constant 532 nm pulsed laser irradiation at 400 mW. Gold nanoparticles with larger initial LSPR wavelengths exhibited faster EC evolution and higher apparent growth rate constants. Among the investigated samples, the 540 nm gold core exhibited the fastest shell growth behavior, followed sequentially by 536 nm, 532 nm, 525 nm, and 520 nm. This trend indicates that larger gold nanoparticles with stronger plasmonic responses facilitate more efficient plasmon-induced catalytic activity during silver shell growth.

• An increase in silver shell growth rate with larger gold nanoparticle diameter was observed under pulsed laser irradiation, which initially appears to contradict the size-dependent trend reported by Baldi and co-workers for spherical nanoparticles. However, their framework attributes decreasing EC rate with size specifically to spherical morphologies, while noting that faceted nanoparticles can exhibit enhanced hot carrier generation at sharp surfaces and corners. Visual inspection of the synthesized nanoparticles suggests that larger nanoparticles deviate from spherical shapes, potentially explaining the observed trend. SEM characterization is ongoing to confirm this hypothesis, but the current evidence points toward shape-driven reversal of the size-dependent EC rate.

With these observations, this work demonstrates that nanosecond pulsed laser excitation provides a highly promising platform for plasmonic catalysis due to its capability to generate extremely high instantaneous electromagnetic fields and enhanced plasmonic excitation within ultrashort timescales. Compared with conventional illumination methods, pulsed laser excitation offers stronger plasmonic enhancement, potentially leading to more efficient catalytic processes and improved control over nanoparticle synthesis.

Furthermore, additional investigations should be conducted in future studies. Time-resolved spectroscopy experiments may provide deeper insight into hot-carrier dynamics and ultrafast energy relaxation processes during pulsed laser excitation. Numerical electromagnetic and thermal simulations such as Finite Element Method (FEM) calculations are also necessary to quantitatively evaluate the contribution of localized photothermal heating during shell growth. In addition, exploring different nanoparticle geometries, shell materials, laser pulse durations, and repetition rates may further improve the understanding and optimization of pulsed laser-induced plasmonic catalytic systems.

Finally, from a broader perspective, the findings of this work open new opportunities for the development of plasmon-driven catalytic systems utilizing pulsed laser excitation for various applications, including photocatalysis, nanoscale synthesis, sensing, energy conversion, and light-assisted chemical reactions